Project Report
(Smart Crop Recommendation System)

Course: Fundamentals of AI and ML

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Abstract

This project, titled "Smart Crop Recommendation System," leverages supervised machine learning to assist in precision agriculture. By analyzing soil chemical properties (Nitrogen, Phosphorus, Potassium) and environmental variables (Temperature, Humidity, pH, and Rainfall), the system provides a data-driven recommendation for the most suitable crop to plant. This eliminates traditional trial-and-error farming, aiming to increase agricultural productivity and sustainability.

Problem Statement & Motivation

The Problem

Agriculture is highly sensitive to environmental factors. In many regions, farmers continue to plant crops based on historical patterns without realizing that soil nutrients deplete over time or that weather patterns have shifted. This lack of data-driven decision-making leads to:

- **Reduced Crop Yield:** Planting crops in unsuitable soil.
- **Resource Waste:** Over-use of fertilizers to compensate for poor crop choice.
- **Economic Risk:** Financial failure for small-scale farmers when a harvest fails.

The AI Solution

The goal of this project was to create a tool that acts as a "Digital Agriculture Consultant." By feeding the model real-time soil data, it can instantly categorize the environment and suggest a crop from a list of 22 varieties (including Rice, Maize, Cotton, Coffee, etc.).

Methodology & Technical Approach

Data Acquisition

The project uses a structured dataset in **CSV (Comma Separated Values)** format. The dataset consists of 2,200 entries. Each entry represents a unique combination of:

- **Nutrients:** N (Nitrogen), P (Phosphorus), K (Potassium) in mg/kg.
- **Climate:** Temperature (°C), Humidity (%), and Rainfall (mm).
- **Soil Chemistry:** pH level (0 to 14).

Machine Learning Pipeline

I implemented the following pipeline using **Python** and the **Scikit-Learn** library:

1. **Data Loading:** Using `pandas` to read and structure the CSV data into a DataFrame.
2. **Feature Selection:** Separating the "Features" (Inputs) from the "Labels" (Target Crop).
3. **Data Splitting:** Using a 80-20 split (80% for training the AI and 20% for testing its accuracy).
4. **Model Training:** I selected the **Random Forest Classifier**.

Why Random Forest?

In my **Fundamentals of AI** course, I learned about various algorithms. I chose Random Forest over a simple Linear Regression or a single Decision Tree because:

- **Non-Linearity:** Soil data doesn't follow a straight line; it has complex patterns.
- **Accuracy:** Random Forest creates an "ensemble" of many decision trees and takes the majority vote, making it highly resistant to errors (overfitting).

Implementation Details

Code Architecture

The project is hosted on GitHub with a clean directory structure:

- **`Code/` Folder:** Contains the core logic and dataset. This ensures that the environment is modular and easy to navigate.
- **`requirements.txt`:** Lists dependencies (`pandas`, `scikit-learn`), ensuring any user can replicate the setup.
- **Interactive CLI:** The Python script uses `input()` functions, allowing a user to interact with the trained model in real-time.

Execution Logic

The code follows a logical flow:

1. Load CSV → 2. Train Model → 3. Wait for User Input → 4. Generate Prediction.

Challenges & Solutions

During the development of this BYOP, I encountered several "real-world" hurdles:

- **Challenge 1: Environment Configuration.** Initially, my VS Code terminal did not recognize the installed libraries. **Solution:** I learned how to manage Python Interpreters and use the terminal to manually install packages via `pip`.
- **Challenge 2: Data Interpretation.** I had to understand the scale of the data. For example, realizing that a pH of 7 is neutral while a pH of 4 is highly acidic helped me validate if the model's output for crops like "Grapes" (which prefer slightly acidic soil) was logical.
- **Challenge 3: Version Control.** Learning how to push code to GitHub and organize it into folders was a new experience that improved my technical documentation skills.

Performance Evaluation & Results

The model was evaluated using the **Accuracy Score** metric. Upon testing, the Random Forest model achieved high accuracy, correctly identifying the crop in nearly every test case.

- **Test Case Result:** When provided with high Nitrogen (90) and high Rainfall (200mm), the model accurately predicted **Rice**.
- **Validation:** The output screenshot included in the GitHub repository proves the system is functional and ready for deployment.

```
PROBLEMS    OUTPUT    DEBUG CONSOLE    TERMINAL    PORTS                                                    Python

PS C:\Users\user\Documents\Python Scripts> & C:\Users\user\AppData\Local\Programs\Python\Python313\python.exe "c:/Users/user/Documents/Python Scripts/raghav"
Data loaded successfully from CSV!

--- Welcome to the Farmer's Assistant ---
Please enter the following soil details:
Nitrogen (N): 22
Phosphorus (P): 10
Potassium (K): 32
Temperature (in °C): 30
Humidity (in %): 20
Soil pH level: 7.4
Average Rainfall (in mm): 52
C:\Users\user\AppData\Local\Programs\Python\Python313\Lib\site-packages\sklearn\utils\validation.py:2691: UserWarning: X does not have valid feature names, but
RandomForestClassifier was fitted with feature names
  warnings.warn(
--------------------------------------
RESULT: You should plant COTTON for the best yield!
--------------------------------------
PS C:\Users\user\Documents\Python Scripts>
```

Final Reflection & Conclusion

This project was a transformative experience in my first year of B.Tech. It taught me that **Machine Learning is not just about code—it is about problem-solving.** By completing this BYOP, I have gained a deeper understanding of:

1. How to structure a professional GitHub repository.
2. The importance of "clean data" in training an AI.
3. How to bridge the gap between a CSV file and a useful, interactive tool.

I believe this system has the potential to be expanded into a mobile app, providing every farmer with an "AI Expert" in their pocket.